

CLOUD COMPUTING

Unit II: Virtualization Technology

Hypervisors, VM Architecture & Types of Virtualization



SAGE
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—BHO PAL—

Unit II – Topics Covered

- What is Virtualization? Definition & Virtual Machine Monitor (VMM)
- Examples and Characteristics of Virtualization
- Hypervisor – Types: Bare Metal (Type 1) and Hosted (Type 2)
- Full Virtualization and Paravirtualization
- Open-Source Hypervisors – Xen & KVM Architecture
- Types of Virtualization – Server, Desktop, Application, Network, Storage, Data
- Case Study



What is Virtualization?

- The process of creating a virtual, rather than physical, version of a resource – such as an OS, server, storage device, or network
- Allows multiple virtual instances to run on a single piece of physical hardware
- A software layer called the hypervisor abstracts and partitions the underlying physical resources
- Each virtual instance operates independently, unaware that it shares hardware with others
- Forms the foundational technology that makes cloud computing possible



Virtual Machine Monitor (VMM)

- The VMM, also called a hypervisor, is the software layer that creates and manages virtual machines
- Sits between the physical hardware and the virtual machines it hosts
- Allocates CPU, memory, storage, and network resources to each virtual machine
- Ensures isolation, so that one VM's activity cannot affect another running on the same host
- Enables multiple operating systems to run concurrently on a single physical machine



Examples of Virtualization

- Running Windows and Linux together on one laptop using VirtualBox or VMware Workstation
- Data centers consolidating many physical servers into virtual machines on fewer physical hosts
- Cloud providers such as AWS, Azure, and GCP using virtualization to offer virtual machines as IaaS
- Application virtualization tools like Citrix delivering software without local installation
- Network virtualization creating multiple virtual networks over shared physical infrastructure



Characteristics of Virtualization

- **Partitioning:** multiple applications and operating systems run side by side on a single physical system
- **Isolation:** each virtual machine is isolated from the host and other VMs, ensuring security and stability
- **Encapsulation:** an entire VM's state and data is stored as a single file, making it easy to copy or move
- **Hardware Independence:** VMs can be migrated between different physical hosts without modification
- **Aggregation:** multiple physical resources can be combined and presented as one larger virtual resource



What is a Hypervisor?

- Also known as a Virtual Machine Monitor (VMM)
- A software or firmware layer that creates and runs virtual machines
- Abstracts physical hardware – CPU, memory, storage – into virtual resources
- Allocates and schedules resources dynamically among multiple virtual machines
- Broadly classified into two categories: Bare Metal (Type 1) and Hosted (Type 2)



Types of Hypervisor

Bare Metal (Type 1)

Runs directly on the physical hardware, with no underlying host OS. Offers better performance and security. Examples: VMware ESXi, Microsoft Hyper-V, Xen.

Hosted (Type 2)

Runs on top of a conventional host operating system, like any other application. Easier to install and use – suited to desktop use. Examples: VMware Workstation, Oracle VirtualBox.



Full Virtualization vs Paravirtualization

Full Virtualization

The guest OS runs unmodified and is unaware it is virtualized; the hypervisor fully emulates the underlying hardware. Offers strong compatibility and isolation.

Paravirtualization

The guest OS is modified to know it is virtualized and communicates directly with the hypervisor through hypercalls. Offers better performance with lower overhead.



Open-Source Hypervisors

- **Xen:** a Type 1 bare-metal hypervisor originally developed at the University of Cambridge, widely used by cloud providers
- **KVM:** Kernel-based Virtual Machine – turns the Linux kernel itself into a Type 1 hypervisor
- **QEMU:** an open-source machine emulator, often paired with KVM for hardware virtualization
- **VirtualBox:** an open-source Type 2 hypervisor, popular for desktop and testing use
- **Benefit:** open-source hypervisors reduce licensing costs and give cloud providers greater flexibility



Xen Architecture

- **Domain 0 (Dom0):** a privileged control domain that manages hardware access and other virtual machines
- **Domain U (DomU):** unprivileged guest domains that run the actual user virtual machines
- **Xen Hypervisor:** a thin software layer running directly on hardware, scheduling CPU and memory among domains
- **Flexibility:** supports both full virtualization and paravirtualization
- **Adoption:** widely used by cloud providers, including early versions of Amazon EC2



KVM Architecture

- Converts the Linux kernel into a Type 1 hypervisor by adding a kernel module (kvm.ko)
- Each virtual machine runs as a regular Linux process, scheduled by the standard Linux kernel scheduler
- Uses QEMU for hardware emulation – devices and I/O – alongside KVM for CPU and memory virtualization
- Leverages hardware-assisted virtualization extensions (Intel VT-x, AMD-V) for near-native performance
- Widely used in open-source cloud platforms such as OpenStack



Types of Virtualization (I)

Server Virtualization

Partitioning a single physical server into multiple isolated virtual servers, each running its own OS.

Desktop Virtualization

Hosting a desktop operating system on a remote server, accessed by end users over a network.

Application Virtualization

Running an application in an isolated environment, decoupled from the underlying operating system.

Network Virtualization

Combining physical network resources into a single, software-based virtual network.



Types of Virtualization (II)

Storage Virtualization

Pooling physical storage from multiple devices into a single, centrally managed virtual storage resource.

Data Virtualization

Providing a unified, real-time view of data drawn from multiple sources, without physically moving or copying it.



Unit II – Case Study & Summary

- **Case Study:** Amazon EC2 uses Xen, and now its custom Nitro Hypervisor, to run many isolated customer VM instances on shared physical servers
- Virtualization abstracts hardware into flexible, isolated virtual resources
- Hypervisors – bare metal (Type 1) and hosted (Type 2) – manage and allocate resources to virtual machines
- Server, desktop, application, network, storage, and data virtualization address different infrastructure needs
- **Next:** Unit III – Cloud Based Services

